TRoss 303 — Acid Bass for Max for Live

TRoss 303 is a TB-303 style bass synthesizer with a built-in 303-style step sequencer, 64-pattern memory, pattern chaining and a random pattern creator. The sound comes from the **Lemus ro-mono-303** engine: a part-by-part circuit model of the original voice board (pitch and slide, VCO, the four-pole transistor ladder filter, envelopes, accent sweep, VCA and output stage), compiled into the device so it sounds the same as the Lemus plugin.



The device panel (preview rendering).

1. Installing

Unzip the download. You get a folder **TRoss 303** containing the device, the panel script and the engine for Mac (*tross303~.mxo*) and Windows (*tross303~.mxe64*). Keep these files together in the same folder, and drag the whole folder into the **User Library** in Live's browser. Then drag **TRoss 303.amxd** onto a MIDI track.

macOS: files downloaded from the web are quarantined, and Live won't load the engine until the quarantine flag is removed. Run this once in Terminal (adjust the path to where the folder is):

```
xattr -dr com.apple.quarantine ~/Music/Ableton/User\ Library/TRoss\ 303
```

Single-file device: to make one self-contained .amxd (for example to share or sell), open the device in the Max editor, press **Freeze Device** in the toolbar, and save. Freezing embeds the panel script and both engine files.

Requirements: Ableton Live 11 or 12 with Max for Live (Max 8.6 or newer), macOS 11+ (Apple Silicon or Intel) or Windows 10/11 64-bit.

2. Sound controls

Control	What it does
TUNING	Fine pitch, about a fifth either way (centre = in tune).
CUT OFF	Filter cutoff of the 4-pole ladder.
RESO	Resonance. High settings give the classic squelch.
ENV MOD	How far the filter envelope sweeps the cutoff.
DECAY	Filter envelope decay time (accented notes always use the short decay, like the original).
ACCENT	How strongly accented steps (and MIDI notes with velocity 100+) hit the filter and VCA.
VOLUME	Output level.
WAVEFORM	Saw or square oscillator: click the switch.
CUT MAX	(Readout at the top.) How high the CUT OFF knob reaches at full (1200-3000 Hz). Real units differ: 2300 Hz is a typical one, lower is darker.
TRANSP	Drag the readout up or down. Transposes the sequencer's notes, in semitones (-24 to +24).
SEQ	Click the SEQ LED to turn the built-in sequencer on or off. With it off, the device is a MIDI-played 303 voice.

Every control is a Live parameter: you can automate it, map it to MIDI or a Push, and it's saved with your set.

3. Playing from MIDI

MIDI notes on the track play the voice too, the way Roland's own TB-03/TD-3 take MIDI: **velocity 100 or more is an accent** (velocity is otherwise ignored), and a note that starts while another is still held **slides** to its pitch without retriggering. Overlap notes in a clip to get slides. While you hold MIDI notes they take over from the sequencer.

TRoss 303 is an instrument, so it doesn't send MIDI out to other tracks (Max for Live instruments can't). To get its patterns as MIDI, use **MIDI EXPORT** (section 8).

4. The sequencer

The sequencer follows Live's transport: press Play in Live and it starts from step 1 of the current pattern (or of the chain), in sync with Live's tempo; press Stop and it stops. Each pattern has up to 16 steps, and every step has five lanes:

Lane	How to edit	Meaning
NOTE	Click and drag up/down. Clicking a rest step makes it a note. Alt-click turns it back into a rest.	C to C' (the 303's 13 keys).
OCT	Click to cycle: none, UP, DN.	One octave up or down.
ACCENT	Click to toggle.	Accented step: louder, brighter, faster filter sweep.
SLIDE	Click to toggle.	Glides from this step into the next one, without retriggering.
TIME	Click to cycle: note (dot), tie (dash), rest (empty).	Tie holds the previous note through this step. A tie after a rest plays nothing.

The **STEP** LEDs above the steps show which step is playing (when the pattern you're editing is the one playing; while a chain plays, the step editor follows the pattern that is sounding). **Click a STEP LED to set the pattern length** to that step (1 to 16); steps past the end are dimmed. The LCD shows the pattern's length as *L16*, *L12* and so on.

5. Pattern memory

Like the TB-303 there are **64 patterns**: 4 groups (**I-IV**) × 8 patterns (**1-8**) × 2 sections (**A/B**). Pick a group, a pattern and a section to select one; the panel shows the pattern you're editing in large letters on the LCD, and **PLAY** with the one playing below it. As on the original, a new selection made while playing takes over when the current pattern reaches its end, so changes stay in time: the LCD shows it as **NEXT**.

The **red LEDs** under the pattern keys 1-8 always show the pattern that is **playing** (when it's in the group and section shown). A pattern you selected that will play next **blinks**. When Live is stopped, the LED shows the selected pattern. The orange LEDs show the group (I-IV) and section (A/B) you're in.

The LCD

Shows	Meaning
I-1A	The pattern you're editing (large).
L16	Its length in steps.
T+3	Its own transpose (see below).
PLAY / STOP / NEXT	What's playing, whether Live is stopped, and a pattern waiting to play next.
CHAIN	Chain information while chaining.

Touch the LCD (click it) and the acid smiley appears. **Click and drag up or down on the LCD** to set the selected pattern's own **transpose**, -12 to +12 semitones (shown as T+3, T-5…). Each pattern keeps its own transpose, so a chain can move a line around without rewriting it. It adds to the global TRANSP control, is saved with the pattern, is copied and pasted with it, and UNDO takes back a whole drag.

Button	What it does
COPY / PASTE	Copy the selected pattern, then paste it into another one.
CLEAR	Clears the selected pattern (all rests, 16 steps). While chain EDIT is on, it clears the chain instead.
UNDO	Undoes the last change to the selected pattern (random, variation, clear, paste, transpose or an edit). Press again to redo.

Every key on the panel goes down visibly while you press it. All 64 patterns and the chain are saved inside your Live set, and each instance of the device has its own.

6. Pattern chaining

A chain plays up to 16 patterns in order and loops (for example I-1A, I-1A, I-1B, I-2A). Press **EDIT** (next to CHAIN) and click pattern buttons to add them to the end of the chain; the current group and section are used, and clicking A or B adds that section; the LCD shows how many entries the chain has and the last one added. Press EDIT again to stop adding. **CHAIN** (the gold key; its LED is lit when on) turns chain playback on or off, and **CLEAR** pressed while EDIT is on empties the chain. With chain playback on, the sequencer plays the chain from its first entry each time Live starts. While the chain plays, the step editor, the LCD and the red pattern LEDs follow the pattern that is sounding, so you can watch (and edit) each one as it comes round.

7. Random pattern creator

The gold **RANDOM** key writes a new acid line into the selected pattern, and the LCD greets it with an acid smiley for a moment. UNDO brings back the previous one.

SHIFT + RANDOM makes a small **variation** of the pattern instead of a new one: 2 to 4 steps change one thing each (a note moves within the scale, an octave flips, an accent or slide comes and goes, a step plays or rests), so the line stays recognisable. Press it again for more variation; UNDO steps back. Variations follow the same SCALE, ROOT and LOCK settings.

The controls next to it shape what it makes:

Control	What it does
SCALE / ROOT	Notes stay in this key. *Acid* is a root-heavy minor-seventh set, the classic sound; *Chromatic* uses every note.
DENSITY	How many steps play rather than rest (downbeats play a little more often, so lines keep a pulse).
ACCENT	Chance of an accent on each step.
SLIDE	Chance of a slide on each step.
OCTAVE	How many steps get an octave UP or DOWN.
LOCK	Click a LOCK key (its LED lights) to keep a lane as it is when re-rolling: for example lock NOTE and OCT and press RANDOM again to re-roll only the rhythm, accents and slides.

8. MIDI export

MIDI EXPORT (bottom right) saves a standard MIDI file (.mid). It exports the **selected pattern**, or the **whole chain** in order when CHAIN is on. A save dialog asks where to put it; the name is suggested for you (for example *TRoss303 I-1A.mid* or *TRoss303 chain.mid*). Drag the file into Live (or any DAW) to get a MIDI clip of the line. The key's LED lights and the LCD says MIDI SAVED when it's done.

The file plays back the same through TRoss 303, a TB-03/TD-3 or any 303 that takes MIDI: accented steps have velocity 127 (others 90), slides are overlapping notes, ties are long notes. Both transposes (TRANSP and the pattern's own) are included, and the file carries Live's current tempo.

9. About, credits and license

Click the **TRoss 303** logo for the About window; click the Gumroad link (on the panel or in the About window) to open the store.

TRoss 303 by Ticco Ross — ticcoross@gmail.com — https://ticcoross.gumroad.com/

The sound engine is **Lemus ro-mono-303** (https://github.com/lemus-dev/lemus, version 0.1.3), used unchanged. Lemus and TRoss 303 are free software under the **GNU Affero General Public License v3**: the complete source code of this device is at https://github.com/Spunkmeyer77/TRoss303. TB-303 is a trademark of Roland Corporation; this product is not affiliated with or endorsed by Roland.

10. Version history

Version	Changes
1.2.0	MIDI export of a pattern or the chain. SHIFT + RANDOM makes a variation. Per-pattern transpose (drag on the LCD). Bigger LCD with larger text and NEXT pattern info. The smiley appears whenever the LCD is touched. Red pattern LEDs follow the playing pattern (the next one blinks). The step editor follows the sounding pattern while a chain plays. Every key shows when it's pressed.
1.1.1	Fix: LOCK keys turned themselves on when the device loaded, so RANDOM changed nothing; SEQ could start switched off.
1.1.0	Silver Classic panel: chicken-head knobs, rubber keys, LEDs, green LCD with acid smiley, gold keys. Pattern length set from the STEP LEDs.
1.0.1	Fix: the panel drawing hid the controls.
1.0.0	First release: Lemus TB-303 voice, 303 sequencer, 64-pattern memory, chain, random creator.